Ananya Anower

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EDUCATION

M.Sc. in Management of Technology (MOT) — CGPA: 3.75/4.00
Bangladesh University of Engineering & Technology (BUET), Bangladesh — April 2023- Present
Thesis Title: Strategic approach to transform medical waste management from linear to circular economy in Bangladesh integrating Artificial Intelligence. **[Thesis Ongoing]**

B.Sc. in Biomedical Engineering (BME) — CGPA: 3.05/4.00
Military Institute of Science & Technology (MIST), Bangladesh — February 2019 – February 2023
Thesis Title: Design and Development of an In-sole Foot Pressure Sensor System for Gait Analysis.

Higher Secondary School Certificate (HSC) — 2018
Rajuk Uttara Model College, Uttara, Dhaka

Secondary School Certificate (SSC) — 2016
Rajuk Uttara Model College, Uttara, Dhaka

EXPERIENCE

Bangladesh University of Engineering and Technology (BUET) - *Research Assistant* — Nov 2023 -Nov 2024
Topic: Strategic approach to facilitate the integration of artificial intelligence in transforming household waste management in Bangladesh from a linear to a circular economy

- Developed a machine learning model for primary image classification that has a high degree of accuracy in classifying images into different groups. The model is being improved for real-time data classification, and its performance is being optimized for useful applications.
- Developing a thorough system to separate garbage from within and outside the home to increase sustainability and efficiency. It is a comprehensive waste management plan and an economic transformation program for Bangladesh with an emphasis on cutting-edge methods and sustainable growth.

Academic Tutor (Online and Offline) — 2020 - Present

- Taught students from classes 9 and 10 (Science- Physics, Chemistry, General and Higher Math & Biology)
- Taught students from classes 11-12 (ICT, General and Higher Math, Physics, Chemistry & Biology)

Evercare Hospital - *Biomedical Engineering Trainee* — 2022 (1st - 31st March)

- Gained direct knowledge of the operation and maintenance of an oxygen gas pipeline system, dialysis machine, patient monitoring system, ECG machine, surgical diathermy, and ultrasound machine calibration.
- Engaged with patients regularly to get insight into their difficulties, which improved my capacity to offer knowledgeable and sympathetic support in the medical context.

GAOTek Inc. - *Digital Marketing Intern* — April 2022 - Sept. 2022

- Attended the intended clients and produced the appropriate Excel documents, guaranteeing precise and well-organized records of customer communications and information.
- Reviewed the organization's CRM, tracking engagement, managing connections with customers, and streamlining the sales process to increase customer happiness and retention.

RESEARCH INTEREST

Cardiovascular Diseases, Phonocardiogram, Neurobehavioral research, Device Designing, Artificial Intelligence, Machine Learning, Instrumentation, Simulation, App Development, Gait Analysis, Technology Management, Biomechanics, Biomaterials

RESEARCH WORK

[1] Hakim, F., Ahmed, J., Anower, A., Chowdhury, M.S., Dutta, A., Ornob, A. (2024). **PCGNet: Convolutional Neural Network to Classify Phonocardiogram Signals – has been accepted for publication in the 4th IEEE International Conference on Robotics, Automation, Artificial Intelligence and Internet of Things**

[2] Development of Household Solid Waste Management Models for Dhaka City to Enhance the Circular Economy in Bangladesh – **has been accepted for publication in Environmental Research & Technology (DergiPark Akademik)**

[3] Currently working on a project on applying BCI for ADHD in children.

PROJECTS

[1] Expert Opinion Methodology Forecasting Mental Health Technology 2024
- forecasted a mental health technology known as Deep TMS (Transcranial Magnetic Stimulation) H-coil system by Brainsway utilizing the expert opinion technique since experts had differing views on using this technology.

[2] Internet of Things Business Model Innovation and the Stage-Gate Process: An Exploratory Analysis 2023
- Explored how industrial business models are impacted by the Industrial Internet of Things (IIoT), emphasizing the value of adaptability and inventiveness.

[3] PCGNet: Convolutional Neural Network to Classify Phonocardiogram Signals 2022
- Built the PCGNet CNN architecture, which uses class balancing strategies to increase accuracy and equity amongst various signal categories.
- Demonstrated PCGNet's potential for early heart problem diagnosis by classifying phonocardiogram signals acquired from a PCG collection device which scored highest among all the projects in our batch.

[4] Design and Simulation of a Flex-Foot Prosthesis using SOLIDWORKS, ANSYS & OpenSim 2022
- Designed a flex-foot using **SolidWorks**, producing an accurate and comprehensive 3D model suited for maximum comfort and performance. Performed extensive **ANSYS** assessments of load, stress, and deformation to guarantee the flex-foot's dependability and longevity in various scenarios. Further simulation was done in **OpenSim** for detailed analysis.

[5] Implementation of Hill's Muscle Model in MATLAB Simulink 2022
- Demonstrated the three-element biomechanical Hill's muscle model, which sheds light on the dynamics and functionality of muscles in many settings. The force-velocity behaviors of the muscles were analyzed using Simulink, which improved our understanding of muscle mechanics in real-time applications and enabled the simulation of numerous physiological scenarios.

[6] Development of a patient monitoring device combining ECG, PPG, and SPO2 sensors. 2022
- Developed a portable, affordable EEG and PPG acquisition system, making cardiovascular monitoring more accessible for clinical and research settings. Categorized arrhythmias using advanced sensors to measure heart rate, enabling timely detection and analysis of irregular heart rhythms for improved patient care.

[7] Arduino-based monitor to Measure Heart Rate & Disease Detection 2021
- Modeled an Arduino simulation that measures heart rate and may detect bradycardia and tachycardia in people. Through real-time data processing, the simulation offers quick feedback on variations in heart rate, allowing for early intervention and encouraging heart health monitoring.

TOEFL SCORE

97 (R 26, L 21, S 24, W 26)

SKILLS

Engineering Software: SolidWorks, Mimics, ANSYS, Kinovea, OpenSim, Proteus Design Suit, EMU8086 Microprocessor Emulator, Latex, Android Studio, Google Colab.
Programming Language: C, Arduino, MATLAB, Python
Other Software Skills: Adobe Illustrator, Photoshop, Adobe Premiere Pro, Filmora, Microsoft Office, Google Suits

AWARDS & ACHIEVEMENTS

- **Best Transformative Business Idea**- IEEE WIE Climate Tech Big Idea Pitch Competition 2023
- **Ministry Of Education Board Scholarship** – Junior School (Talent Pool- secured 1st place in Dhaka board), Primary School (Talent Pool- secured 5th place in Dhaka board)
- **Certificate of Internship (**GAOTek Inc.), **Certificate of Training in Event Management (**GAOTek Inc.)

EXTRACURRICULAR ACTIVITIES

Seminar Series Program on Technology-Focused Entrepreneurial Thought Leaders (BUET) Feb 2024
- Anchored the program and won praise for showing remarkable presence and public speaking
- helped put the program together, showcasing strong leadership where our primary objective was increasing connectivity with Bangladesh's successful entrepreneur leaders so we could get their input on fresh ideas and hear about their inspirational journeys.

MIST Einthoven Club - ***Creative Design Director*** May 2022 - Feb 2023
- Designed for the club using Adobe Illustrator, Photoshop and Video editors. Carried out several departmental events with success
- Assessed the junior designers' work and offered helpful criticism and direction to help them develop their abilities. Made sure the finished designs fulfilled the project's goals and requirements, encouraging teamwork and learning.

IEEE MIST Student Branch – *Webmaster* June 2022 - Feb 2023
- Pulled IEEE-organized programs onto our campus, successfully planning and publicizing activities to increase student involvement. As an involved member, I was a key player in organizing their efforts, ensuring students had access to important materials and networking opportunities.

REFERENCES

Dr. Iftekhar Uddin Bhuiyan
Associate Professor
Institute of Appropriate Technology,
Bangladesh University of Engineering and Technology (BUET)
Email: iftekhar@iat.buet.ac.bd

Lt Col Md. Maruf Hasan, Ph.D, EME
Associate Professor
Department of Biomedical Engineering, Tower-4,
Military Institute of Science & Technology (MIST)
Email: maruf.hasan@bme.mist.ac.bd